PHISHING EMAIL ANALYSIS

Identifying Malicious Emails & Indicators of Compromise

SOC Portfolio Project · March 2026

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Certification	Certified SOC Analyst (CSA) – EC-Council
Project Type	Phishing Email Analysis – Portfolio Project
Platform	TryHackMe – Phishing Emails 1 & Phishing Emails in Action
Tools Used	TryHackMe · Thunderbird · CyberChef · Mozilla VM
Date	March 2026
Status	Completed ✓

1. Project Overview

This project demonstrates the analysis of phishing emails to identify malicious indicators such as spoofed senders, suspicious URLs, malicious attachments, and credential harvesting techniques. Two TryHackMe rooms were completed to gain hands-on experience with real phishing email samples.

Key skills demonstrated in this project:

- Analyzing email headers to identify spoofed senders and suspicious origins
- Identifying malicious URLs and defanging them for safe investigation
- Recognizing fake login pages used for credential harvesting
- Identifying malicious attachments and embedded tracking pixels
- Understanding phishing techniques such as urgency and brand impersonation
- Mapping phishing tactics to the MITRE ATT&CK; framework

2. TryHackMe Rooms Completed

Room	Description	Difficulty	Status
Phishing Emails 1	Email structure, headers, delivery protocols and analysis	Easy	100% ✓
Phishing Emails in Action	Real phishing email samples and indicator analysis	Easy	100% ✓

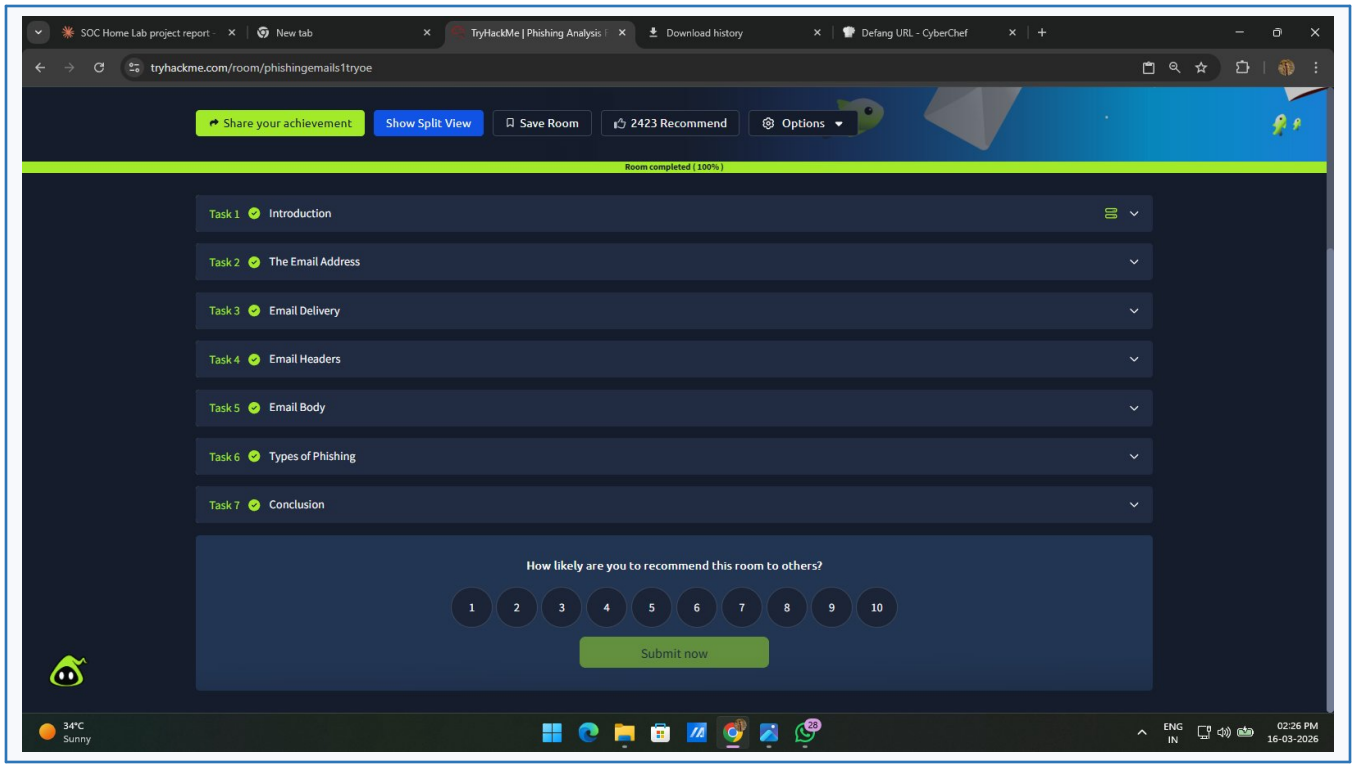

Figure 1: TryHackMe – Phishing Emails 1 completed with 100% progress (All 7 tasks complete)

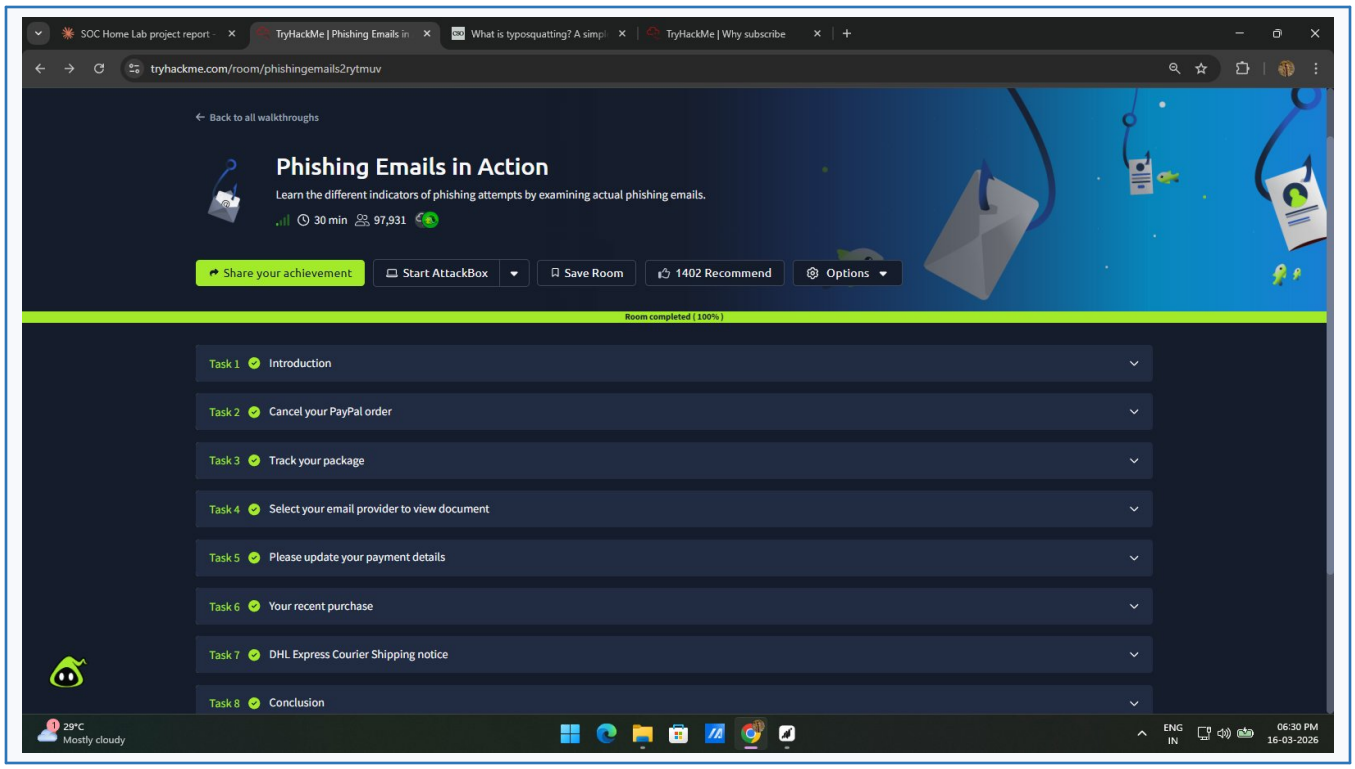

Figure 2: TryHackMe – Phishing Emails in Action completed with 100% progress (All 8 tasks complete)

3. Email Header Analysis

Email headers contain critical information that SOC analysts use to investigate phishing emails. Key fields examined during this investigation include the From, Sender, Reply-To, and X-Originating-IP fields.

3.1 Key Email Header Fields

Header Field	Purpose	Phishing Indicator
From	Display name of sender	Can be spoofed to impersonate trusted entities
Reply-To	Address replies are sent to	Different from From = suspicious
X-Originating-IP	IP address email was sent from	Investigate via arin.net for origin
Smtp.mailfrom	Actual sending domain	Mismatch with From = spoofing attempt
Return-Path	Same as Reply-To	Check for domain mismatch

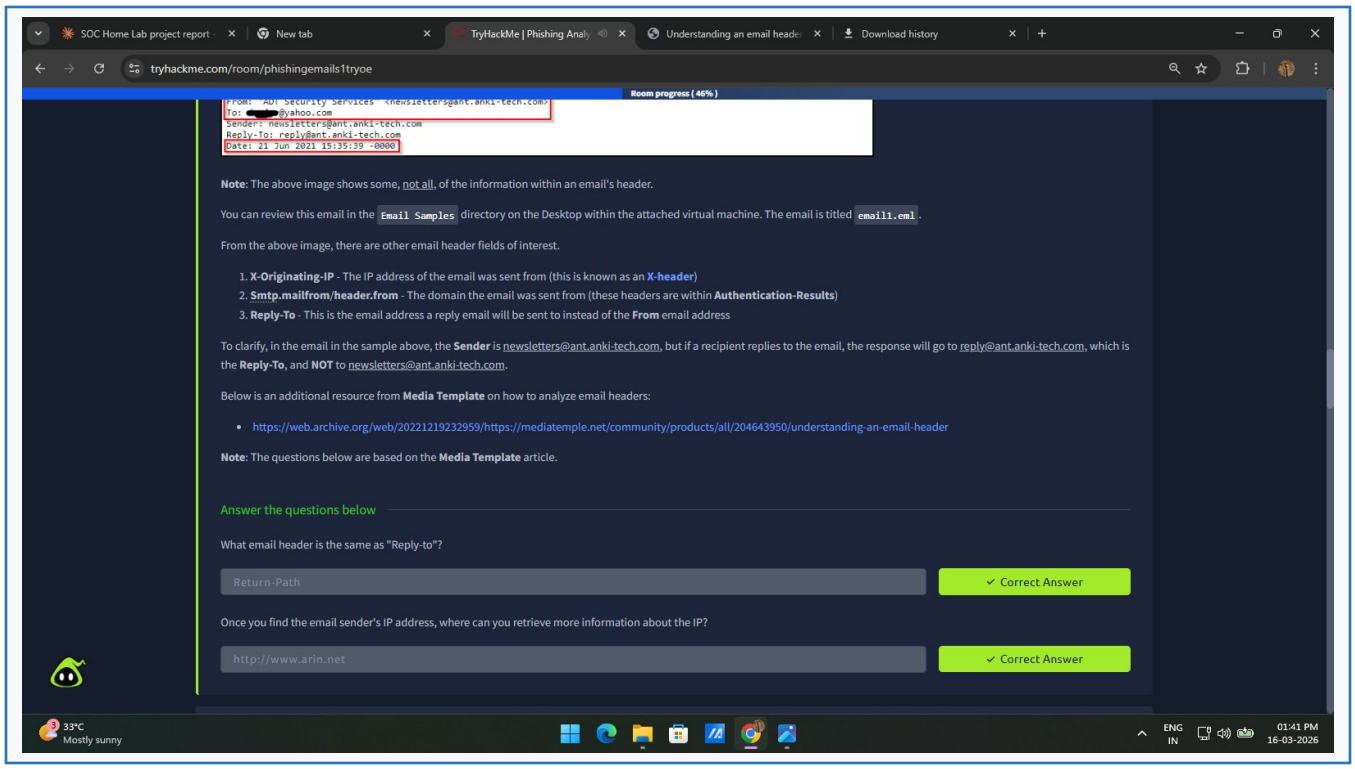

Figure 2: Email header analysis showing spoofed sender – newsletters@ant.anki-tech.com masquerading as ADI Security Services

The email header revealed that the sender was **newsletters@ant.anki-tech.com** with a Reply-To address of **reply@ant.anki-tech.com**, both different from the display name **ADI Security Services** — a clear indicator of email spoofing.

4. Phishing Email Sample Analysis

A real phishing email sample was analyzed in the TryHackMe virtual machine using Mozilla Thunderbird. The email was designed to impersonate Home Depot and trick the victim into clicking a malicious link.

4.1 Email Details

Field	Value
From (Display)	Thank you! Home Depot
Sender Email	support@teckbe.com
To	alexa@yahoo.com
Subject	Order Placed : Your Order ID OD2321657089291 Placed Successfully
Malicious URL	hxxp[://]t[.]teckbe[.]com
Technique	Brand Impersonation – Home Depot

MITRE ATT&CK | T1566.002 – Phishing: Spearphishing Link

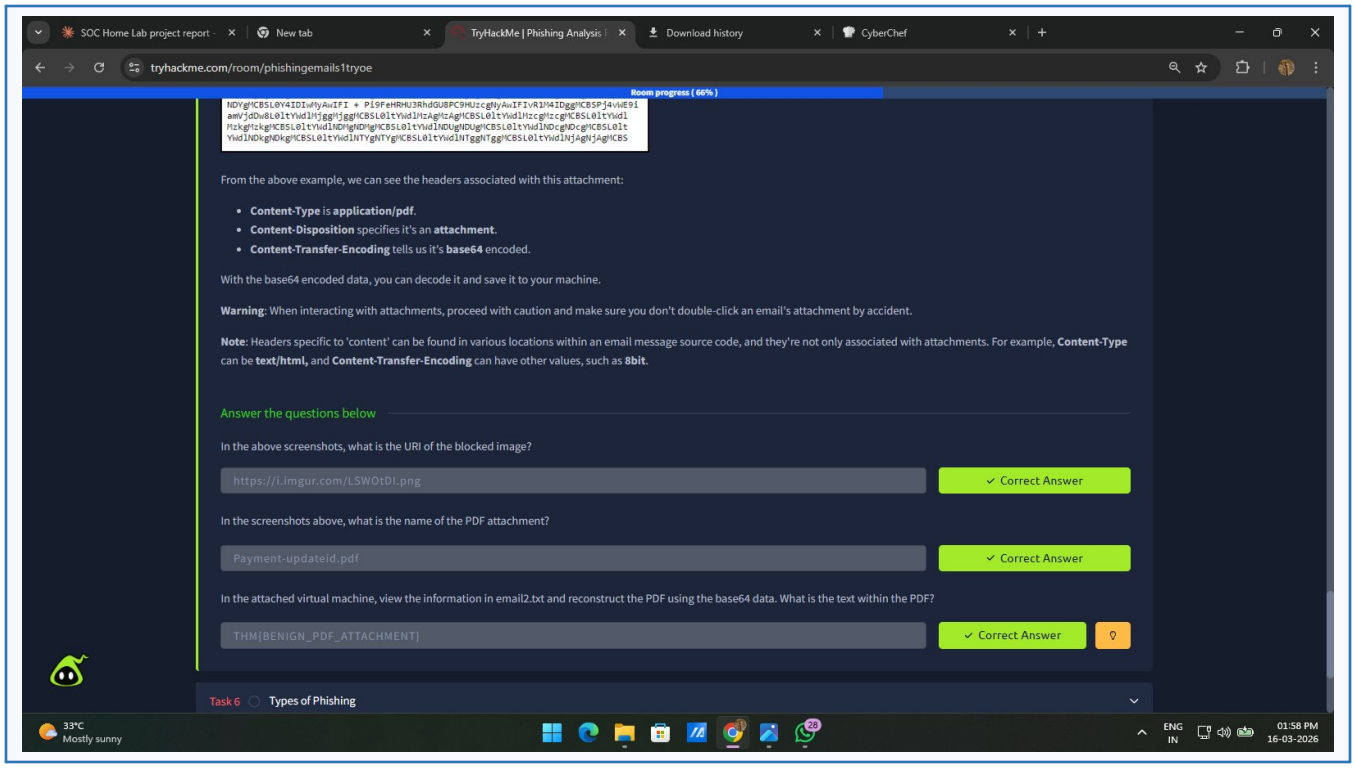

Figure 3: Email body analysis showing malicious PDF attachment Payment-updateid.pdf with base64 encoded content

5. Malicious URL & Domain Analysis

Malicious URLs found in phishing emails must be defanged before investigation to prevent accidental clicks. CyberChef was used to defang URLs by replacing special characters.

5.1 URL Defanging

URLs are defanged by replacing characters to make them unclickable:

```
Original: http://devret.xyz/4833mt11254939vf6888zq22032si1269du1508rr

Defanged: hxxp[://]devret[.]xyz/...
```

5.2 Suspicious URLs Identified

URL (Defanged)	Risk	Finding
hxxp[://]devret[.]xyz	High	Associated with malware distribution
hxxp[://]t[.]teckbe[.]com	High	Fake Home Depot redirect URL
hxxps[://]i[.]imgur[.]com/...	Medium	Blocked tracking image

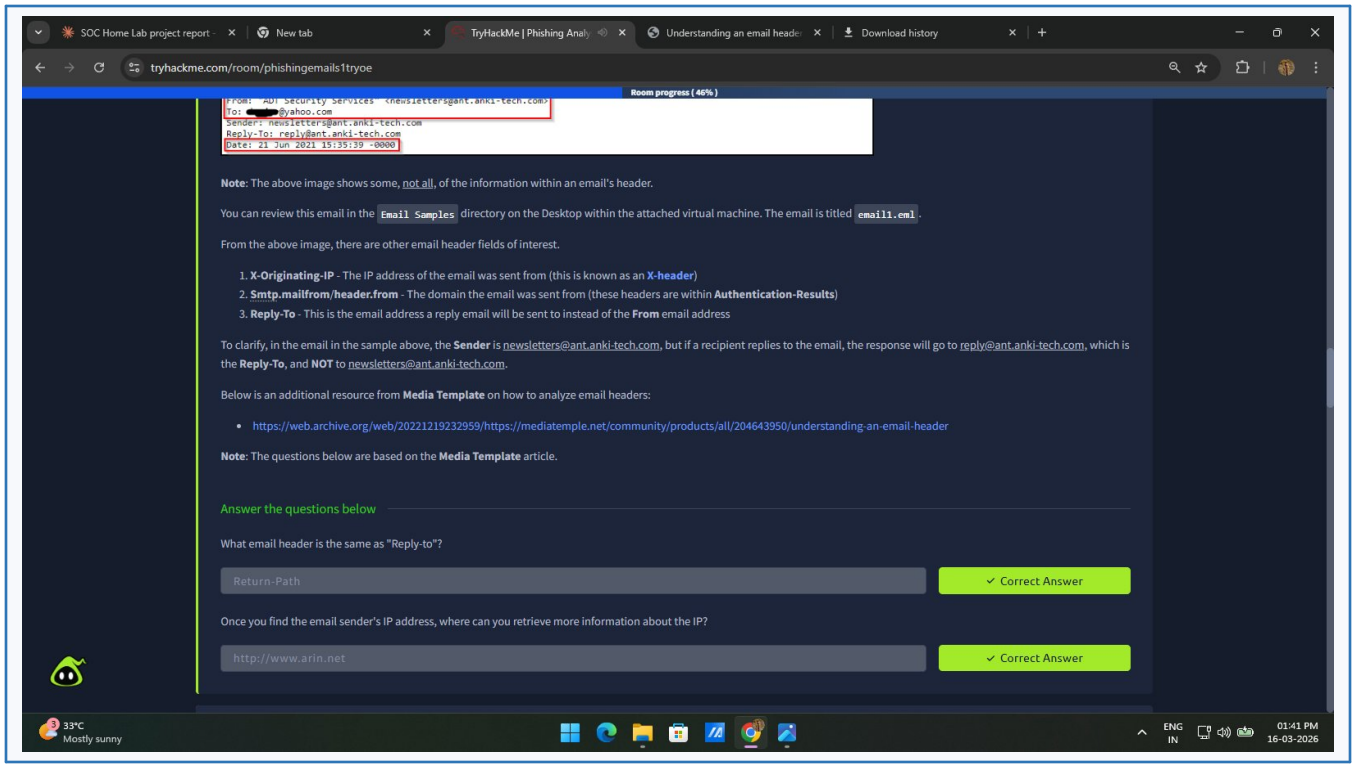

Figure 4: Email header investigation showing spoofed sender and suspicious reply-to address

6. Credential Harvesting – Fake Login Page

One phishing email led to a fake Microsoft Outlook login page designed to steal user credentials. The page was hosted on a malicious domain and showed an 'Invalid Credentials' error regardless of what was entered — confirming the credentials were being sent to the attacker's server.

MITRE ATT&CK | T1056.003 – Input Capture: Web Portal Capture

Key indicators identified:

- Fake Microsoft Outlook login page hosted on malicious domain
- Invalid Credentials shown regardless of input — credentials sent to attacker
- Email impersonated OneDrive, Adobe, and Citrix brands to appear legitimate
- Urgency technique used to pressure victim into acting quickly

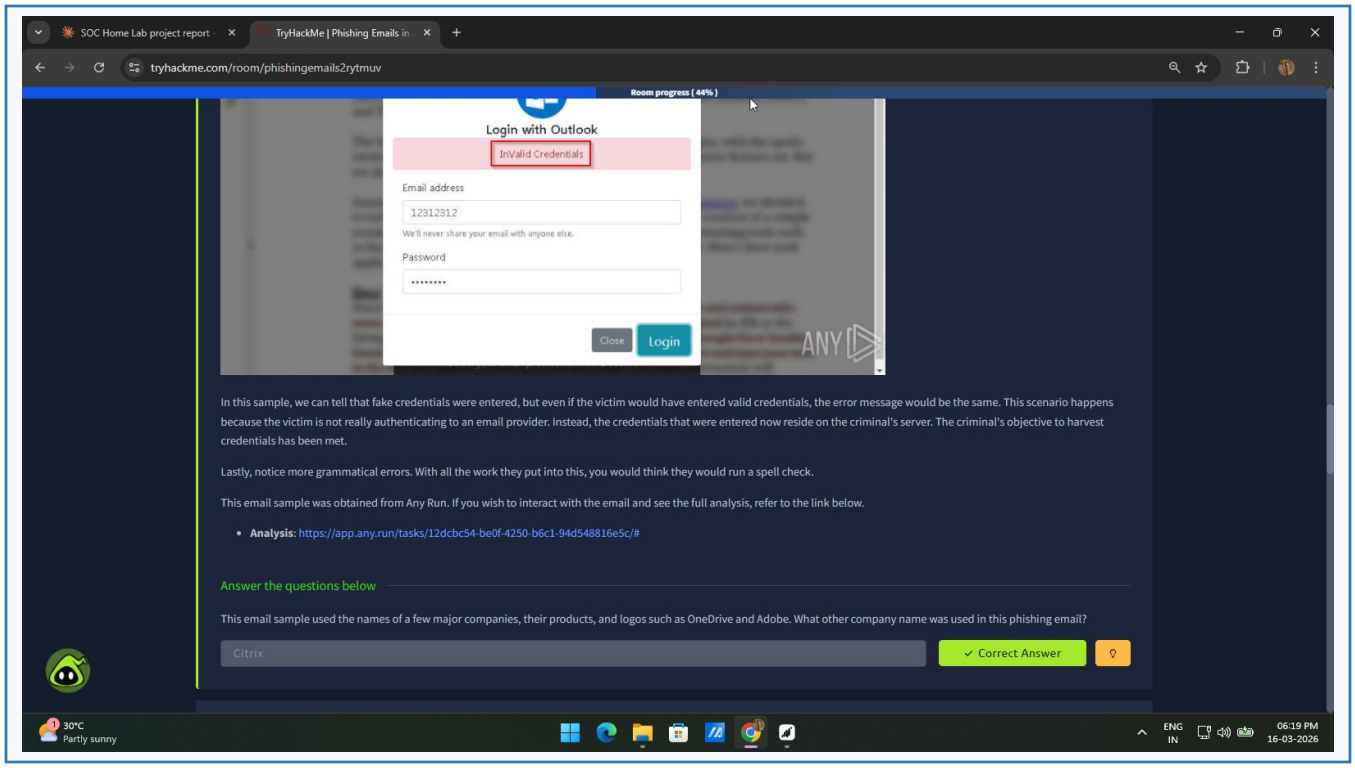

Figure 5: Fake Outlook login page used to harvest credentials – shows Invalid Credentials after submission

7. Findings

#	Finding	Indicator	MITRE ATT&CK;
1	Email Spoofing	Sender/Reply-To mismatch with display name	T1566.001
2	Malicious URL	hxxp[://]devret[.]xyz – malware domain	T1566.002
3	Brand Impersonation	Home Depot, Microsoft, Adobe, Citrix logos used	T1566
4	Credential Harvesting	Fake Outlook login page stealing credentials	T1056.003
5	Tracking Pixel	Tracking.png embedded to track email opens	T1598
6	Malicious Attachment	PDF attachment Payment-updateid.pdf	T1566.001

8. Conclusion

This project successfully demonstrated how a SOC analyst identifies and investigates phishing emails using TryHackMe lab environments and real phishing email samples.

Multiple phishing techniques were identified including email spoofing, malicious URLs, fake login pages for credential harvesting, and malicious attachments. Each technique was mapped to the MITRE ATT&CK; framework.

Skills demonstrated:

- Analyzing email headers to identify spoofed senders
- Defanging malicious URLs using CyberChef for safe investigation
- Identifying fake login pages used for credential harvesting
- Recognizing phishing techniques such as brand impersonation and urgency
- Mapping phishing indicators to MITRE ATT&CK; framework

This project reflects real SOC analyst workflows when handling phishing alerts and demonstrates readiness for entry-level cybersecurity roles.

9. Tools & References

- TryHackMe – tryhackme.com
- CyberChef – gchq.github.io/CyberChef
- Mozilla Thunderbird
- MITRE ATT&CK; Framework – attack.mitre.org
- EC-Council Certified SOC Analyst (CSA) Courseware